Mahmud Hamda

Field Service Engineer /

Full-Stack Developer &

Cybersecurity Professional Profile

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Technically minded Field Service Engineer, Full-Stack Developer, and CompTIA Security+–certified Cybersecurity Professional with experience resolving complex biomedical equipment issues, leading service operations, and building secure, intelligent websites. Develops responsive, end-to-end web applications with React.js, JavaScript (ES6+), HTML5, CSS3, Bootstrap 5, Node.js, Express.js, MySQL, and custom REST APIs, using Git and GitHub for version control and deployment. Applies a security-first mindset through code audits, secure data handling, access control, cryptography, threat detection, and network security to harden web environments, protect critical user data, and improve resilience against modern threats. Enhances applications with AI capabilities including custom chatbots, semantic search, retrieval-augmented generation (RAG), interactive AI coaches, automated workflows, and contextual intelligence. Proven ability to lead and collaborate with teams, maintain accurate documentation and inventory controls, communicate proactively with customers, and deliver reliable, compliant solutions that advance service quality and business objectives.

Areas of Expertise

Biomedical Field Service Engineering

- Site Preparation & Equipment Installation
- Commissioning & On-Job Training
- Recording, Reporting & Documentation
- Medical Equipment Technology Assessment
- Maintenance, Repair & Calibration
- Asset Management
- Decommissioning & Disposal
- Troubleshooting
- Inventory Management
- Operation & Safety
- Evaluation & Monitoring

Full-Stack Web Development

- React.js, JavaScript, HTML5 & CSS3
- Node.js, Express.js & REST APIs
- MySQL, SQL & Database Design
- AI integration
- Git, GitHub & Deployment

Cybersecurity

- Threat, risks and Vulnerability
- Incident response
- Security frameworks and controls
- Security hardening
- Programing languages (Python, SQL)
- Linux Command-line
- Logs and Network analyzing tools (SIEM, TCPDUMP, Wireshark)

Technical Skills

Cybersecurity: CompTIA Security+, SIEM, IDS/IPS, log analysis, security monitoring, threat detection, incident response, vulnerability assessment, remediation, IAM, access control, firewalls, network security, secure data handling, cryptography, Linux, Python, and SQL.

Full-Stack Development: React.js, JavaScript (ES6+), HTML5, CSS3, Bootstrap 5, Node.js, Express.js, REST APIs, MySQL, database design, Git, GitHub, deployment, secure coding, RAG, semantic search, and AI chatbot integration.

Biomedical Engineering: medical equipment installation, commissioning, diagnostics, troubleshooting, repair, calibration, preventive maintenance (PM/PPM), safety testing, CMMS, asset inventory, lifecycle management, SOPs, regulatory compliance, and technical documentation.

Additional Tools: Microsoft Office, Google Docs, C++, C#, MATLAB, Arduino, Cisco, Oracle, and Proteus.

Professional Experience

Self, Denver, USA **2023 – Present**
AI-Powered Full Stack Developer

Build and deploy responsive, intelligent full-stack web applications using React.js, JavaScript, HTML5/CSS3, Bootstrap 5, Node.js, Express.js, MySQL, and REST APIs. Integrate AI chatbots, semantic search, RAG, and interactive AI tools, with Git and GitHub for version control. Apply CompTIA Security+ expertise to audit code, secure data, enforce access controls and cryptography, detect threats, and harden applications and networks against modern web risks.

- Built and deployed Netflix, Apple website, and ChatGPT clones with semantic search and Retrieval-Augmented Generation (RAG).
- CompTIA Security+ principles Applied to audit code, protect data, manage access controls, implement cryptography, detect threats, and harden networks.

Shaggar City Health Office, Addis Ababa, Ethiopia **2017 – 2023**
Biomedical Field Service Engineer

Optimized medical equipment functionality through troubleshooting, maintenance, installation, and calibration across clustered hospitals in the city. Authored comprehensive documentation defining protocols, policies, and standards for medical equipment maintenance, ensuring consistency and compliance. Streamlined equipment operations by creating comprehensive Standard Operating Procedures (SOP) and adhering to established standards and policies. Recorded and managed inventory, history files, and maintenance logs for medical equipment, ensuring accurate and up-to-date documentation.

- Directed a team of eight Biomedical Engineers and Technicians, ensuring high-performance and efficient operations.
- Promoted to oversee multiple hospitals by driving strategic management across the cluster.
- Reduced yearly average costs by 8M ETB through effective collaboration with Biomedicals in medical equipment lifecycle management, at Shaggar City (former Oromia Special Zone Surrounding Finfinnee) Health Office.
- Accomplished 95% completion rate for medical equipment installations.
- Elevated maintenance success to 93% by implementing proactive strategies, reducing downtime significantly.
- Ensured 99% PPM compliance through rigorous scheduling and streamlined execution across all hospital locations.
- Revamped equipment management by refining record-keeping systems, boosting documentation accuracy and reporting efficiency.
- Developed automated inventory system, enhancing tracking efficiency and overall equipment management.

Chancho Hospital, Addis Ababa, Ethiopia **2016 – 2017**
Biomedical Equipment Technician

Resolved complex issues and performed maintenance and installation of medical equipment, ensuring optimal functionality within the hospital. Oversaw asset inventory management and prioritized classification based on risk levels. Logged, tracked, and generated reports for asset history records within the hospital.

- Accomplished 98% completion rate for medical equipment installations.
- Drove 97% success in medical equipment maintenance, ensuring consistent functionality and reduced downtime.
- Achieved 98% compliance in planned preventive maintenance (PPM) by streamlining hospital maintenance workflows.
- Built software to generate unique Inventory Code Numbers (ICNs), improving asset tracking.
- Optimized equipment data management, significantly improving record accuracy and reporting efficiency.
- Distinguished as the top Biomedical professional for exceptional medical equipment management, documentation and repair proficiency.
- Entrusted with expanded oversight of four hospitals under Shaggar City Health Office, driving operational excellence.

Education

AI-Powered Full Stack Development

- Evangadi, Inc, 2026
- Developed responsive full-stack applications with React.js, Node.js, MySQL, REST APIs, AI chatbots, semantic search, and RAG.

CompTIA Security+ ce Certification

CompTIA, 2024–2025

- Validated expertise in threat management, access control, cryptography, network security, risk management, and incident response.

Bachelor of Science in Biomedical Engineering, Instrumentation & Medical Imaging

Addis Ababa University, Ethiopia, 2016

- Educational credentials evaluated by WES as equivalent to a U.S. Bachelor's degree.

Certifications

AI-Powered Full Stack Developer | Evangadi INC (2026)

- Developed responsive full-stack applications using React.js, Node.js, MySQL, REST APIs, AI chatbots, semantic search, and Retrieval-Augmented Generation (RAG).

CompTIA Security+ SY0-701 Certification (2024 – 2027)

- Successfully completed the requirements to be recognized as CompTIA Security+ Certified

Google Cybersecurity Certification (2024)

- Proficient in cybersecurity tools including Python, Linux, SQL, SIEM, and IDS, with expertise in identifying and mitigating cyber threats and vulnerabilities.

Cybersecurity in Healthcare (Hospitals and Care Centers) Certification | Erasmus University Rotterdam (2023 – 2024)

- Identify cybersecurity risks, threats, and vulnerabilities targeting medical data as well as techniques to mitigate them.

Key Projects

- deployed Netflix-clone, Apple website-clone, and ChatGPT clones : Jan 2026
- Added semantic search and Retrieval-Augmented Generation (RAG) in projects: Jan 2026
- Evangadi Forum: Jan 2026
- Basic Network Scanner: 2025
- Automated Inventory Code Number (ICN) generator: Jan, 2017
- Automated Medical Equipment Database: Jun, 2020
- Remote Monitoring of Patients' Vital Signs: Temperature and Heartbeat Using Proteus Software: Sep, 2015 – Jan, 2016
- Remote Monitoring of Patient's Vital Signs Using GSM Technology: Nurse Assistant: Jan, 2016 – Jun, 2016